

Curvature and Torsion in the Continuum Theory of Lattice Defects

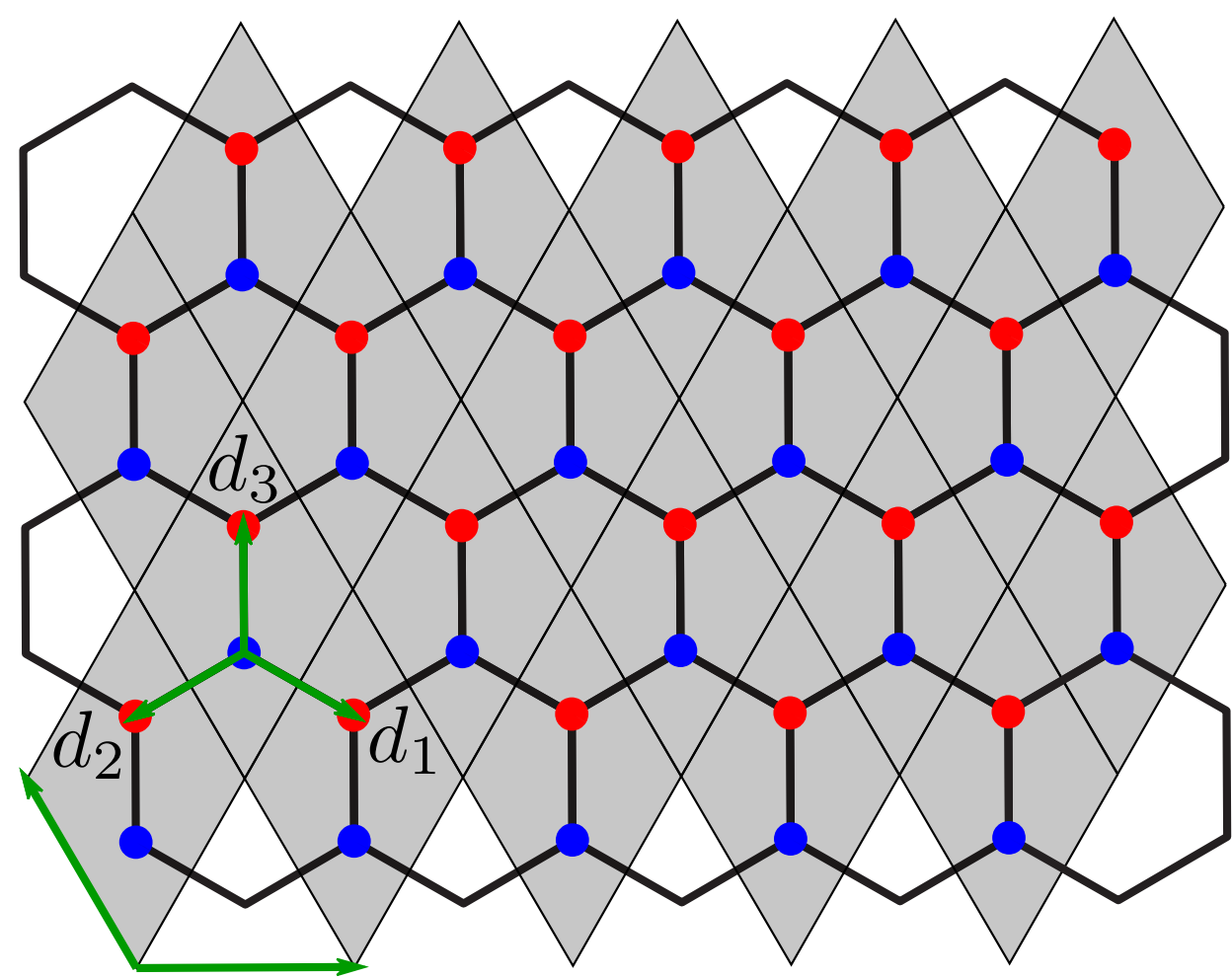
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Continuum Theory - Graphene ^[1]



Tight-binding Hamiltonian

$$H = \sum_{\vec{r}} \sum_{j=1}^3 (a_{\vec{r}}^{\dagger} b_{\vec{r}+\vec{d}_j} + b_{\vec{r}+\vec{d}_j}^{\dagger} a_{\vec{r}})$$

Fourier transform

$$i\partial_t \vec{\psi}(\vec{k}) = \mathbf{h}(\vec{k}) \vec{\psi}(\vec{k})$$

$$\mathbf{h}(\vec{k}) = \begin{pmatrix} 0 & f(\vec{k}) \\ f^*(\vec{k}) & 0 \end{pmatrix}, \quad f(\vec{k}) = \sum_{j=1}^3 e^{i\vec{k} \cdot \vec{d}_j}$$

Linear dispersion around Dirac points

$$\vec{k} = \vec{K}^{(s)} + \delta\vec{k} \implies \mathbf{h}(\vec{k}) \approx -\frac{3}{2} \delta\vec{k} \cdot \vec{\sigma}$$

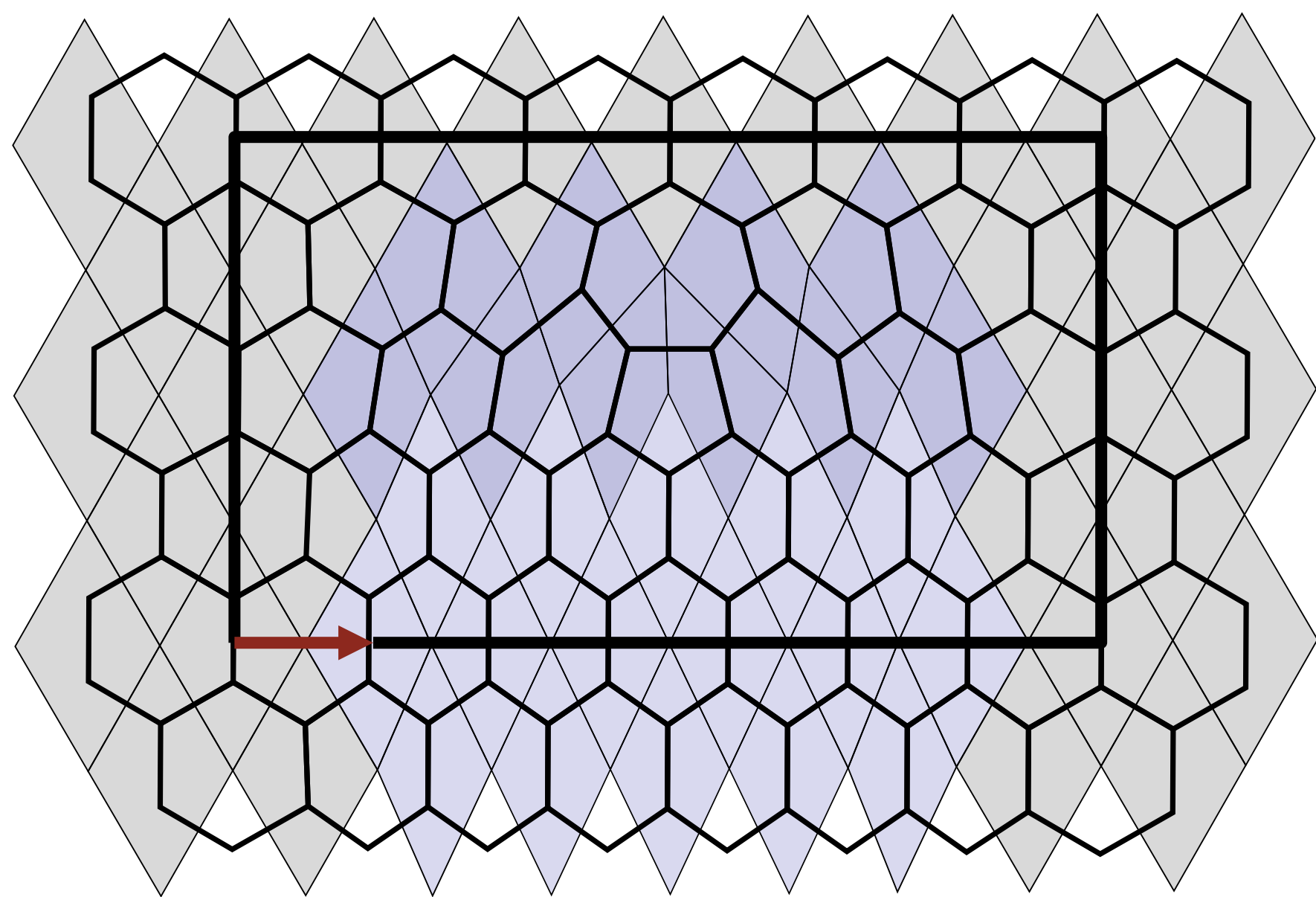
Continuum limit

$$i\partial_t \vec{\phi}(\vec{r}) = i c \vec{\sigma}^i \partial_i \vec{\phi}(\vec{r}), \quad \rho = \vec{\phi}^{\dagger} \vec{\phi}, \quad \vec{j} = -c \vec{\phi}^{\dagger} \vec{\sigma} \vec{\phi}$$

$$\Downarrow$$

$$i\gamma^{\mu} \partial_{\mu} \phi = 0, \quad \gamma^{\mu} = \sigma^{\mu} \sigma^3, \quad \mu \in \{0, 1, 2\}$$

Burgers vector in 5-7 defects ^[2]



One 5-7 defect causes a lattice dislocation which can be described by the Burgers vector $\vec{b} = \sqrt{3} \vec{u}_x$

Torsion ^[3]

In standard GR the torsion is assumed to vanish. It is defined as the antisymmetric part of the connection:

$$T^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\mu\nu} - \Gamma^{\lambda}_{\nu\mu}$$

Any general affine connection can be decomposed into a symmetric part, the contorsion and the disformation:

$$\Gamma^{\alpha}_{\mu\nu} = \{\alpha_{\mu\nu}\} + K^{\alpha}_{\mu\nu} + L^{\alpha}_{\mu\nu}$$

Here we assume **metric-compatibility**:

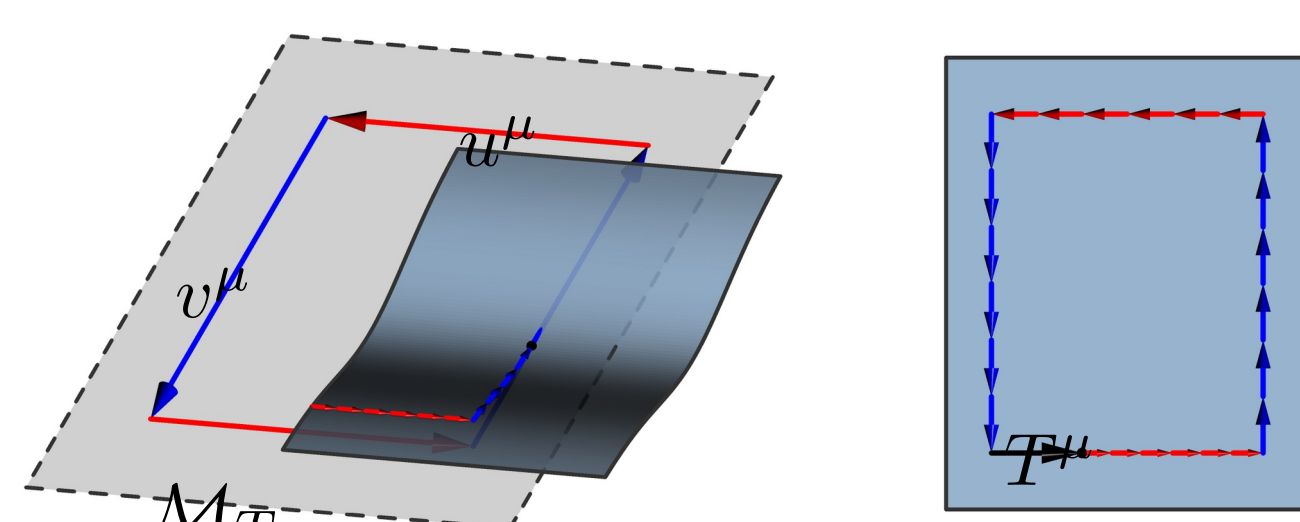
$$\nabla_{\lambda} g_{\mu\nu} = 0 \Leftrightarrow L^{\alpha}_{\mu\nu} = 0$$

The **contorsion tensor** is build from the torsion:

$$K^{\alpha}_{\mu\nu} = \frac{1}{2} (T^{\alpha}_{\mu\nu} - T_{\mu\nu}^{\alpha} + T_{\nu}^{\alpha\mu})$$

Torsion can be thought of as the displacement vector $T^{\mu}(u, v)$ resulting from rolling the tangent plane along a parallelogram:

$$T^{\mu}(u, v) = u^{\lambda} v^{\nu} T^{\mu}_{\nu\lambda}$$

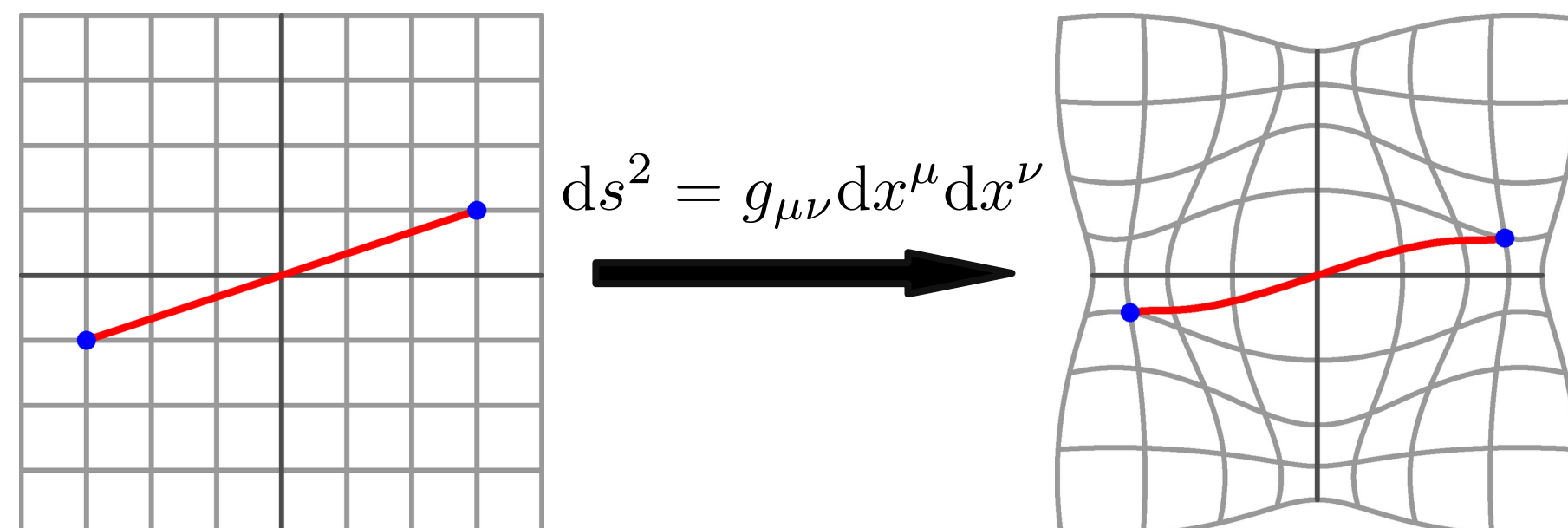


Riemannian Geometry ^[3]

Riemannian Manifold $\mathcal{M}$ equipped with metric $g_{\mu\nu}$ and connection $\Gamma^{\lambda}_{\mu\nu}$.

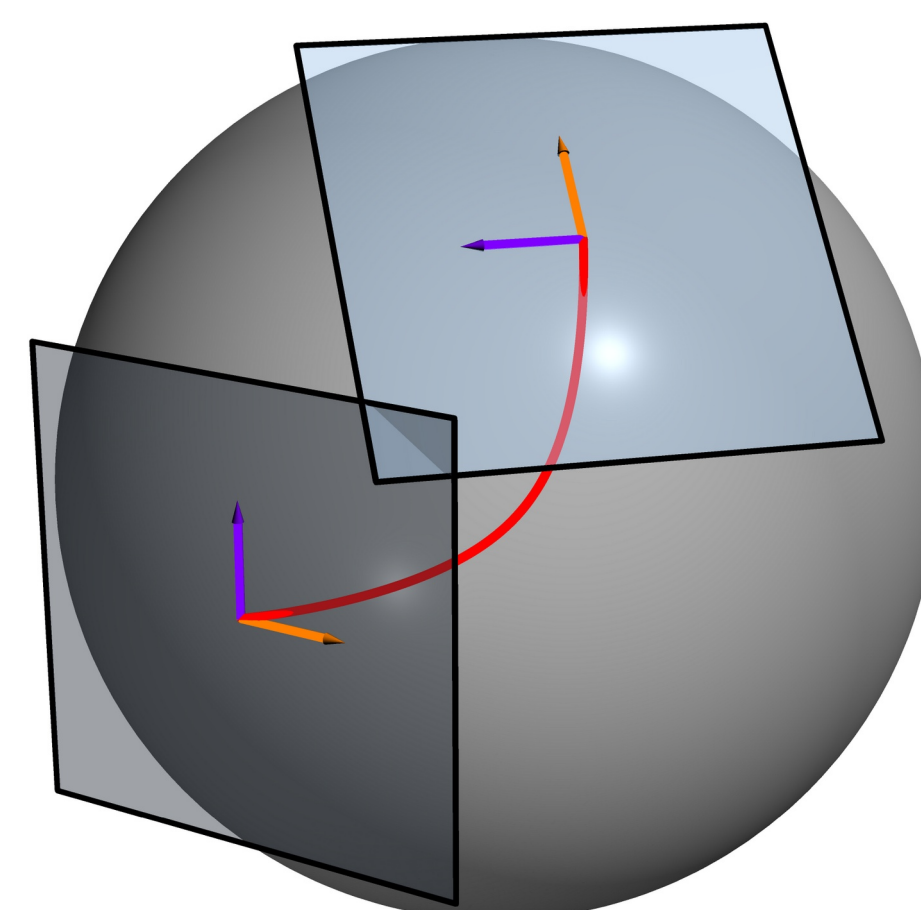
Metric

Defines distances and angles.



Connection

Connects nearby **tangent spaces**, which allows the definition of a derivative on **vector fields** which take on values in the tangent space.



The unique torsion-free, metric compatible connection is called the **Levi-Civita connection**:

$$\{\lambda_{\mu\nu}\} = \frac{1}{2} g^{\lambda\rho} (\partial_{\nu} g_{\rho\mu} + \partial_{\mu} g_{\rho\nu} - \partial_{\rho} g_{\mu\nu})$$

Covariant derivative

Takes into account the structure of the manifold and transforms like a tensor.

$$\nabla_{\mu} v^{\nu} = \partial_{\mu} v^{\nu} + \Gamma^{\nu}_{\mu\lambda} v^{\lambda}$$

Autoparallels

A curve whose tangent vector is parallel-transported along itself.

$$u^{\mu} \nabla_{\mu} u^{\alpha} = 0$$

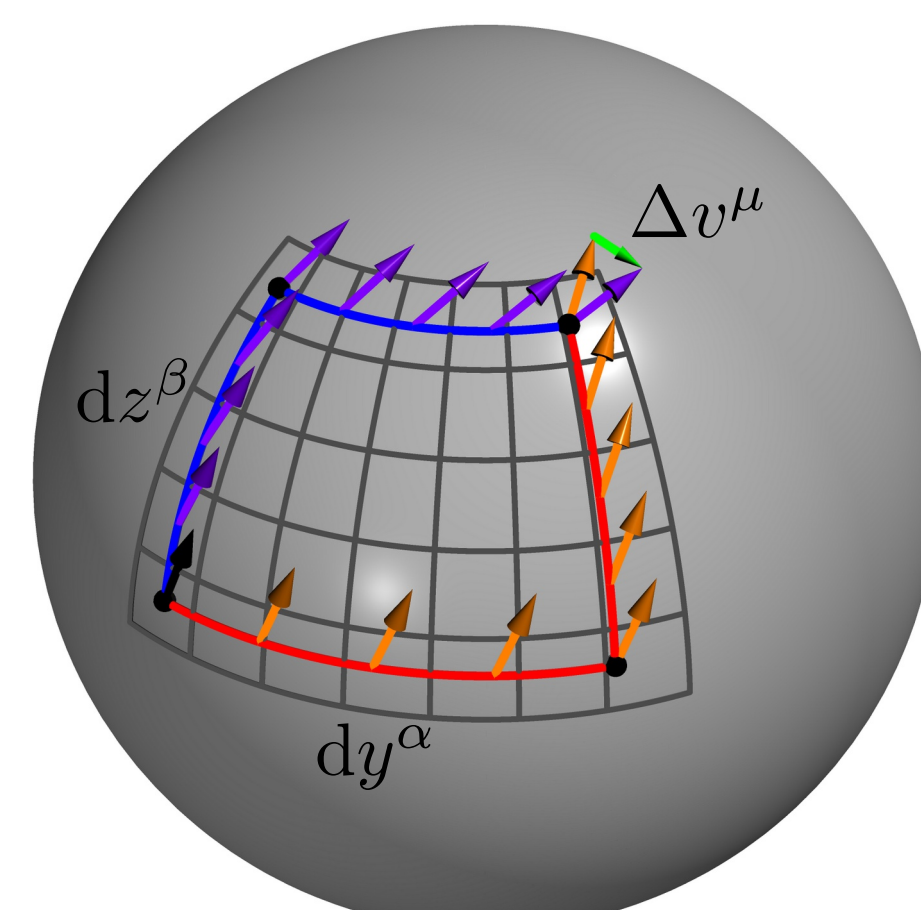
Curvature

The Riemann curvature tensor describes how parallel transport of vectors depends on the path.

$$\Delta v^{\mu} = R_{\alpha\beta}{}^{\mu}{}_{\nu} v^{\nu} dy^{\alpha} dz^{\beta}$$

It can be expressed in terms of the connection:

$$R_{\alpha\beta}{}^{\mu}{}_{\nu} = \partial_{\alpha} \Gamma^{\mu}_{\beta\nu} - \partial_{\beta} \Gamma^{\mu}_{\alpha\nu} + \Gamma^{\mu}_{\alpha\lambda} \Gamma^{\lambda}_{\beta\nu} - \Gamma^{\mu}_{\beta\lambda} \Gamma^{\lambda}_{\alpha\nu}$$



Metric $\leftrightarrow$ torsion mapping in 2D

In 2D we propose a mapping between a **classical** Manifold $\mathcal{M}_g(g_{\mu\nu}, \{\lambda_{\mu\nu}\})$ and **purely torsionful** Manifold $\mathcal{M}_T(\delta_{ij}, K^k_{ij})$.

In 2D we can always choose coordinates in which the metric is $g_{\mu\nu} = \delta_{\mu\nu} e^{2\phi(x, y)}$.

Also the torsion in 2D can be written using a single covector $T^k_{ij} = b^k \varepsilon_{ij}$

We demand equal **curvature** of both manifolds:

$$R[\mathcal{M}_g] = R[\mathcal{M}_T],$$

This leads to the following solution for the torsion vector b^i as a function of the metric ϕ and an arbitrary scalar field χ :

$$b^i = \varepsilon_{ik} \partial_k \phi + \partial_i \chi$$

For $\chi = 0$ the **autoparallels** of both manifolds are equivalent up to a reparameterization. It can be shown, that the χ field corresponds to local rotations of the frame field.

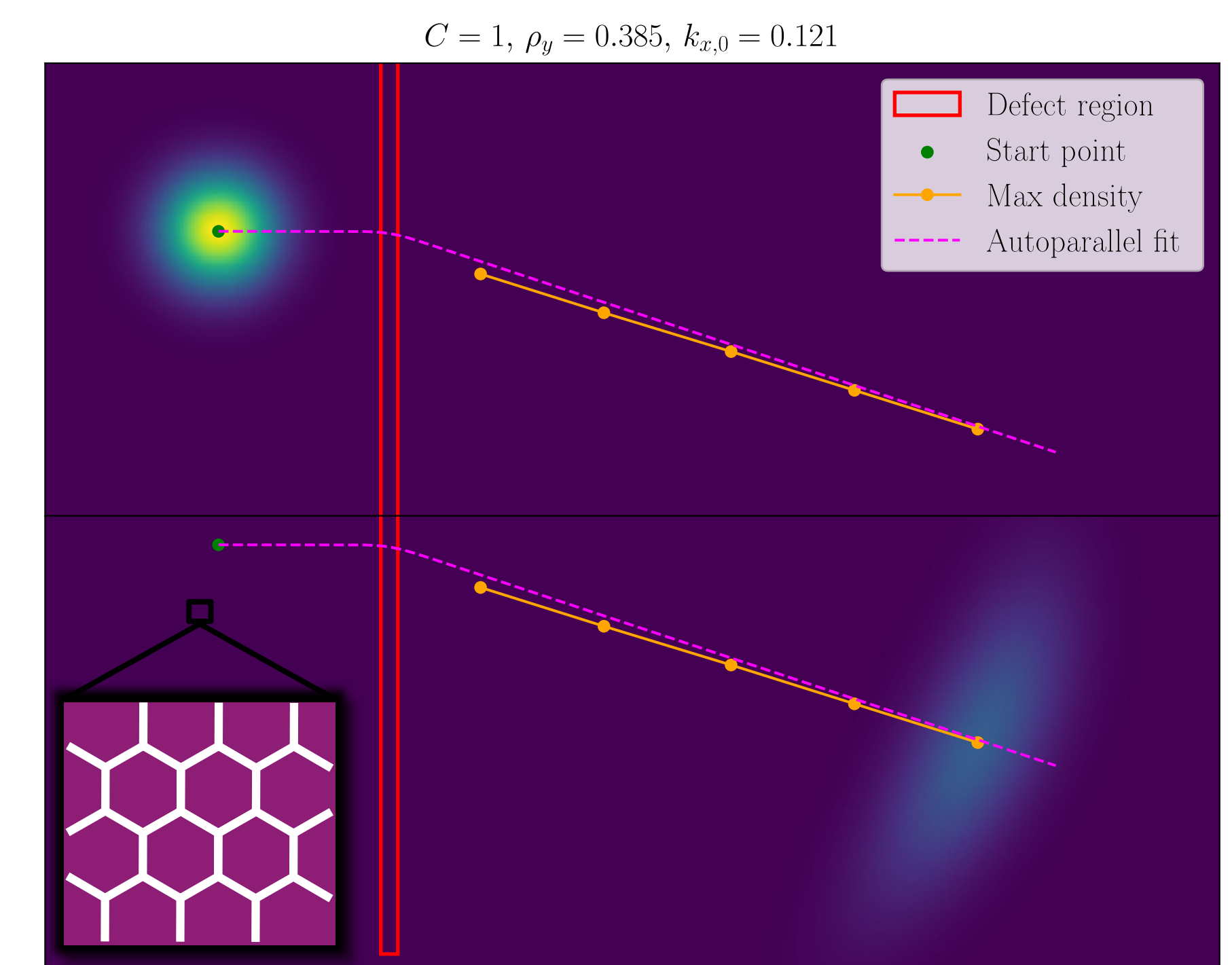
Conclusion

Except for distances and the problem of different tangent spaces a physical mapping between torsion and metric is possible.

Tight-binding numerics

Use a **vector representation** containing the wavefunction at the discrete lattice sites. The Hamiltonian is a **sparse Matrix** acting on this vector to generate the timeevolution:

$$i\partial_t \vec{\psi} = H \vec{\psi} \implies \vec{\psi}(t) = e^{-iHt} \vec{\psi}_0$$



Torsion model

Take $\mathcal{M}_T$ with $b^1(x)$ and $b^2 \equiv 0$. The autoparallel equation becomes:

$$\ddot{x} = -\dot{x}\dot{y}b^1, \quad \ddot{y} = \dot{x}^2 b^1$$

Since $\frac{d}{dt}(\dot{x}^2 + \dot{y}^2) = 0$ we can substitute:

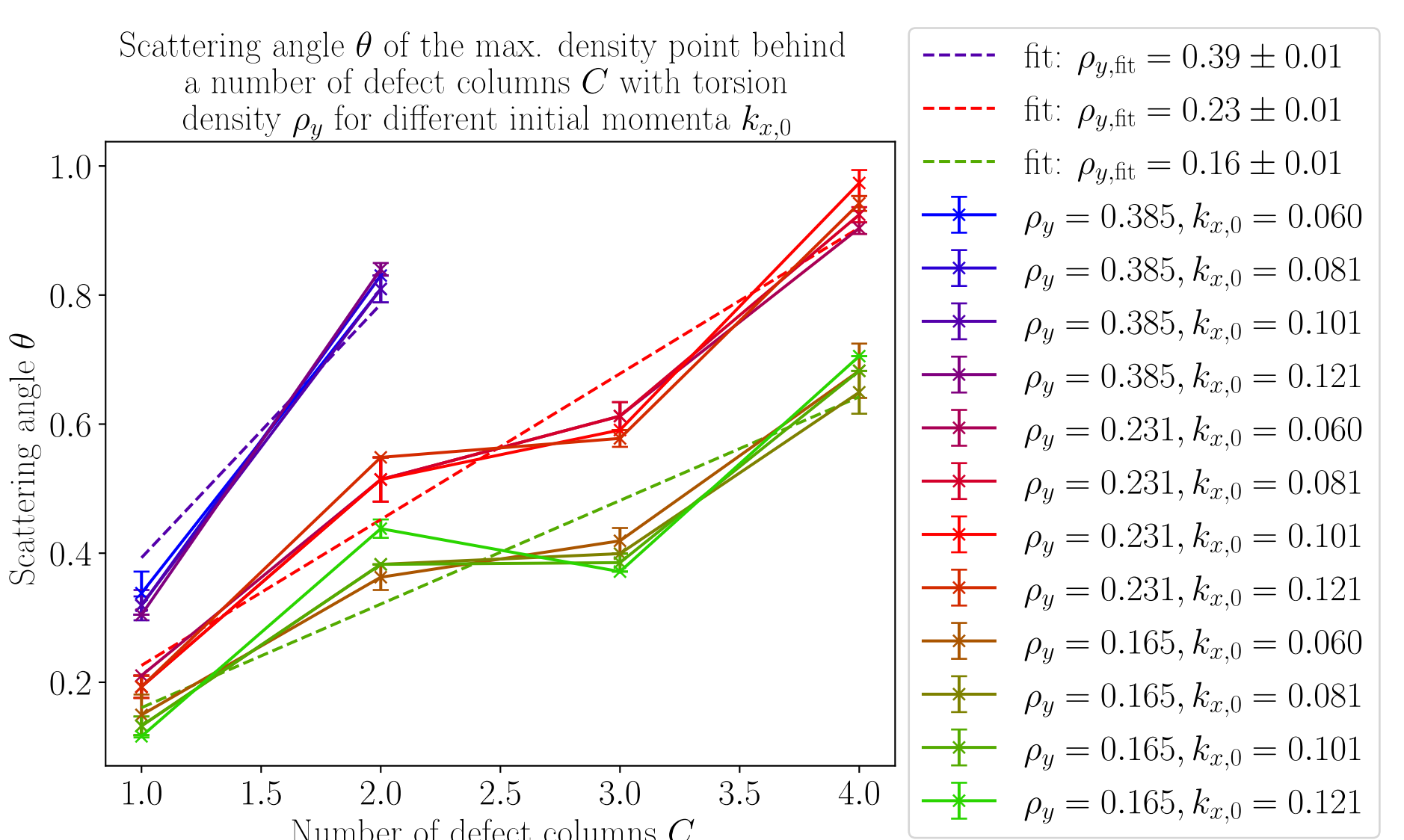
$$\dot{x} = \cos(\theta), \quad \dot{y} = \sin(\theta)$$

The autoparallel equation becomes $\dot{\theta} = b^1(x) \cos(\theta)$. Eliminating the time dependence yields $\frac{d\theta}{dx} = b^1(x)$. We take $\dot{x}(x=0) = 1, \dot{y}(x=0) = 0$. With this we can calculate the scattering angle of the torsion model:

$$\theta_T = \int_0^{\infty} b^1(x) dx$$

Interpreting torsion as a Burgers vector density with distance Δy between defects gives:

$$\rho_y = |\vec{b}|/\Delta y \approx \frac{\iint b^1 dx dy}{\Delta y} = \int b^1 dx = \theta_T$$



Outlook

The classical Dirac equation obtained from continuum theory of graphene can be generalized to describe **curved graphene** by simple generalization to a non-flat geometry ^[1]. The problem is, that this mechanism only works for a torsionless geometry. For a geometry with torsion, which is used to model the scattering with autoparallels, the Lagrangian needs to be constructed in a way that preserves **hermiticity** ^[4]. The question is whether the dynamics resulting from this Lagrangian can describe the tight-binding results.

References

- [1] Stegmann T, Szpak N. Graphene current paths. *NJP* **18**, 053016 (2016).
- [2] Bhatt MD, Kim H, Kim G. Defects in graphene review. *RSC Adv.* **12**, 21520 (2022).
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- [4] de Juan F, Cortijo A, Vozmediano MAH. Dislocations and torsion in graphene. *Nucl. Phys. B* **828**, 625 (2010).